Foundation I: From Source to Manifestation

Accessible Algebras from a Source, Its Noise and a Readable Seed

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October 2026

Abstract

A source, its noise and a physical reader supply different constraints on the structure in which a world can be read. We separate finite measurement permissions from the ideal algebra generated by their effects. A partially mixed record probe generates the same pointer algebra at every nonzero gain, although its sharp-test error approaches one half as the gain vanishes. Repeated fresh nondemolition probes give a finite instrument-level approximation to the sharp measurement, including old references; this supplies a resource-sensitive bridge to finite tasks.

For a stationary source with a faithful state, state-preserving noise has a modular-invariant fixed algebra and a unique state-preserving expectation. Its Cesàro projection need not be a long-time limit of individual trajectories. A stable seed has a modular hull inside that fixed algebra. If the ambient source has positive translations with only the vacuum invariant, every nonscalar hull closed under forward translations inherits a proper Cut on its vacuum space. A minimal compact symmetry identifies any hull containing the symmetry-fixed algebra with the fixed algebra of its seed stabilizer, provided the closure maps preserve it. In the SU(2) level-one net, a positive log-bump stress seed generates the Virasoro half-line algebra under modular evolution alone and its one-sided orbit forms a standard Cut on the Virasoro vacuum space. One bounded current seed generates the current algebra under modular flow alone; its character distinguishes the stabilizer up to conjugacy. Axis dephasing leaves three hulls for the stated stress/current seeds, and weak symmetry breaking determines their passive lifetimes. Stress excitations have the same relative entropy across these hulls, whereas current excitations distinguish them. The finite permission bridge and the net-theoretic Cut construction answer different parts of the source problem: the first determines which ideal distinctions can be used, the second which geometric structure an admitted seed supports. The ambient net, couplings and preparation permissions remain inputs; forward translations are unnecessary for the stated log-bump seeds.

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1 Introduction

Consciousness I [2] works with an accessible algebra A inside a global algebra M with vacuum Ω. When A contains a Cut, a subalgebra stable under the modular evolution in one direction only, the translations of a chiral conformal net are computed from modular data, and for a standard Cut the net and the modular data determine each other. In the SU(2) level-one current net the same paper finds that the algebra fixed by an explicit rotation noise, the $c = 1$ Virasoro algebra of a half-line, coincides with the algebra generated from one bounded stress observable by dilations and forward translations. For a positive log-coordinate bump with monotone ends, the construction needs only modular evolution: its one-sided orbit forms a proper standard Cut on the Virasoro vacuum space, and its full modular hull is the Virasoro half-line algebra. The origin of the ambient structure and of physically available seed operations remains open; the present paper asks which accessible algebra a source and seed select.

Foundation I is the source end of that chain. Its question is what an actual source, physical access to it and persistent organization fix on the way to an accessible algebra, and what they leave free. The HAFF papers supply part of the answer in finite dimension. Paper A [3] shows that two channels can generate the same algebra and still resolve different things, so that one state and one reconstruction rule give different geometries. Paper B [4] derives the exact fixed algebra in the commuting-coupling class and the unique largest block algebra passing its stated window criteria. Block-constant recording couplings expose central labels, while protected relations, classical sector correlations and record-conditioned formation have different roles. Its collective recording and Hamiltonian-link constructions show how actual process resources create relations; the noise graph alone does not determine all correlations. Paper E [6] separates what a probe can implement, what the dynamics leaves fixed and what fragments record, and computes all three in one collision model. For its Gaussian linear-response task with quadratic loss and an unrestricted rank-k projector menu, T-DOME II [8] identifies a top-k posterior-task eigenspace as an optimal reading subspace. A restricted physical instrument menu need not realize that optimizer. What these results do not say is how a seed obtained from access grows into an algebra that can carry geometry, when the result is proper, how it depends on the seed, and whether different seeds give different geometries. HAFF-D lists this as the missing selection step: with a faithful ambient state the ambient algebra itself passes every modular and semigroup stability test, and a nontrivial selection needs a seed and a minimal hull [5].

The objects are as follows. A *source* is a von Neumann algebra $\mathcal{A}_U$ with a cyclic and separating vector Ω, its state ω, and a semigroup $\{\mathcal{E}_t\}$ of normal unital completely positive maps describing the environment in the Heisenberg picture. The modular group σ_s of ω is fixed by the source and needs no further input; for a thermal state it is the time evolution in units of the inverse temperature, and for the vacuum of a wedge it is the boost group. *Access* supplies actual effects through physical probes; a reader may keep a set of them as a *seed*. Algebraic functional calculus then describes the ideal distinctions encoded by that seed. It is a further closure, not an additional measurement permission. The *stable hull* of the seed is the smallest von Neumann algebra that contains it and is closed under the modular flow and environmental dynamics. When an ambient positive translation structure is available, we also study its closure under forward translations. These are distinct closures: translation stability must be imposed or proved before a Cut can be inherited.

The argument proceeds from physical access to persistent structure:

1. Section 3 gives the finite pointer-instrument bridge, including old references and adaptive task error.

2. Sections 4–5 derive the state-preserving expectation and modular stable hull from stationarity, and separate this projection from dynamical relaxation.

3. Section 6 proves proper Cut inheritance for translation-stable nonscalar hulls with unique invariant vacuum.

4. Section 7 identifies the classified hull by its seed stabilizer. Current and positive log-bump stress seeds admit modular-only generation; characters distinguish stabilizers and symmetry-breaking noise fixes passive lifetimes.

5. Sections 8–9 separate ideal seed closure from finite coding budgets and distinguish seed-independent stress response from seed-dependent current response.

The ambient algebra, the coupling and the task that fixes the foreground remain inputs. Positive translations are supplied for the inheritance theorem, but not as a separate closure operation for the stated modular-only seed constructions. Section 10 identifies which further inputs the conjectures would remove. The finite-dimensional statements are checked by scripts described in Appendix B.

2 Source, access, seed and hull

Definition 2.1 (Source)**.** A *source* is a quadruple $(\mathcal{H}_U, \mathcal{A}_U, \Omega, \{\mathcal{E}_t\}_{t\geq 0})$: a von Neumann algebra $\mathcal{A}_U$ on $\mathcal{H}_U$, a vector Ω cyclic and separating for $\mathcal{A}_U$, and a semigroup of normal unital completely positive maps $\mathcal{E}_t : \mathcal{A}_U \to \mathcal{A}_U$, pointwise continuous in the σ-weak topology. We write $\omega = \langle \Omega, \cdot\, \Omega\rangle$, Δ, J for the modular operator and conjugation of $(\mathcal{A}_U, \Omega)$, $\sigma_s = \mathrm{Ad}\,\Delta^{is}$ for the modular group, and

$$\mathrm{Fix}(\mathcal{E}) = \{x \in \mathcal{A}_U : \mathcal{E}_t(x) = x \text{ for all } t \geq 0\}.$$

The source is *stationary* if $\omega \circ \mathcal{E}_t = \omega$ for all t. An *ambient positive translation structure* (a Borchers pair with $\mathcal{A}_U$) is a strongly continuous unitary group $U(a) = e^{iaP}$ with $P \geq 0$, $U(a)\Omega = \Omega$ and $U(a)\mathcal{A}_U U(a)^* \subseteq \mathcal{A}_U$ for $a \geq 0$.

The ambient positive translation structure is the structure of a half-line algebra of a chiral net in its vacuum, or of a wedge algebra with a lightlike translation. By Borchers' theorem it satisfies $\Delta^{it}U(a)\Delta^{-it} = U(e^{-2\pi t}a)$ and $JU(a)J = U(-a)$ [12], and $U(1)\mathcal{A}_U U(1)^* \subset \mathcal{A}_U$ is then a half-sided modular inclusion whose translation group, in the sense of Wiesbrock's theorem, is U [31, 10]. We use it only where it is stated.

Access and seed. Paper E of HAFF describes access by a context: allowed probe channels $\mathcal{R}_\alpha^* : \mathcal{B}(\mathcal{K}_\alpha) \to \mathcal{A}_U$, whose images of probe effects are the implementable effects Eff, the environmental dynamics with its fixed set, and a record algebra [6]. Paper B identifies central record labels for its block-constant recording couplings, not for every possible environment instrument [4]. A seed $S \subseteq \mathcal{A}_U$ is a set of bounded self-adjoint observables. A *physical effect seed* is selected from the actual probe effects and retained under a declared budget (Section 8); an *ideal spectral seed* may additionally contain bounded Borel functions of those effects. Both enter the same mathematical hull, but the latter need not be implementable by the former probe menu. A seed is *stable* if $S \subseteq \mathrm{Fix}(\mathcal{E})$. The hull definitions do not depend on the mechanism selecting the seed. Section 3 supplies a nonempty physical bridge and exhibits the information lost by identifying permissions with this ideal closure.

Definition 2.2 (Stable hull)**.** Let Γ be a family of normal maps of $\mathcal{A}_U$ into itself that contains the modular group, $\{\sigma_s\}_{s\in\mathbb{R}} \subseteq \Gamma$. A von Neumann subalgebra $\mathcal{N} \subseteq \mathcal{A}_U$ is Γ-*admissible* if $\gamma(\mathcal{N}) \subseteq \mathcal{N}$ for every $\gamma \in \Gamma$. The Γ-*hull* of a seed S is

$$\mathrm{Hull}_\Gamma(S) = \bigcap\{\mathcal{N} : \mathcal{N}\ \Gamma\text{-admissible},\ S \subseteq \mathcal{N}\}.$$

We use three families: $\Gamma_\sigma = \{\sigma_s\}$ (the *modular hull*), $\Gamma_2 = \{\sigma_s\} \cup \{\mathcal{E}_t\}$ (the *stable hull*), and, in the presence of an ambient positive translation structure, $\Gamma_3 = \Gamma_2 \cup \{\mathrm{Ad}\,U(a)\}_{a\geq 0}$ (the *translation-stable hull*). Existence of an ambient positive translation structure alone does not identify the Γ_2- and Γ_3-hulls.

Since $\sigma_{-s} \in \Gamma$, an admissible algebra satisfies $\sigma_s(\mathcal{N}) = \mathcal{N}$. For a seed of exact pointer observables realising a foreground frame, Hull_{Γ_2} is the minimal modular–decoherence hull of that frame. The (σ, U)-hull of Consciousness I is Hull_{Γ_3} without the noise maps [2]. HAFF-D calls the Γ_2-admissible algebras modular-stable geometric candidates and notes that the class contains the ambient algebra, so that a selection needs a seed [5].

2.1 Two realizations of the source–access interface

The finite and chiral examples share the preceding definitions, but supply different outputs. In HAFF-B's commuting-coupling model, the semigroup selects the fixed algebra $\bigoplus_a \mathcal{B}(H_a)$ and its central projections P_a. A faithful stationary density matrix is block diagonal. Thus its modular flow fixes every P_a, and

$$\mathrm{Hull}_{\Gamma_2}(\{P_a\}) = W^*(\{P_a\}) = Z(\mathrm{Fix}(\mathcal{E})).$$

This is an instance of Proposition 5.3: the process selects sector distinctions, while the actual probe menu determines their reading accuracy. When controlled nondemolition probes of these projections are permitted, Section 3 gives a finite-cost approximation to their sharp instrument. Neither the central labels nor their repetition construct tensor-product sites or a nontrivial half-sided modular inclusion. This is the finite structural input available to HAFF-A, not an identification of its correlation-graph vertices with these labels.

The chiral realization starts with a net and a vacuum. A noncentral stress or current seed then generates an ideal algebra under its modular flow; symmetry classifies the resulting hulls. A one-sided orbit or an inherited translation inclusion produces a Cut *inside* the appropriate vacuum representation. Proper half-sided inclusion and standardness have separate hypotheses (Section 6); cyclicity on the ambient Hilbert space cannot be assumed. This construction supplies C-I with selected algebras and geometric subnets, without deriving the ambient net from the finite blocks.

The distinction makes the missing connection precise. A passage from finite process-selected structures to a chiral net would have to specify a compatible family of local algebras, states and dynamics, and prove that its limit has the required modular inclusion. A sequence of larger fixed algebras alone supplies none of those relations. The present theorems instead make each end testable: finite protocols must reproduce the claimed instruments, and a proposed net realization must satisfy the hull and Cut conditions.

3 From a physical effect to a finite task

A von Neumann algebra closes under bounded spectral functions, products and operator limits. Probe permissions need not close under these operations. Even the linear inverse of a noisy

effect may require a negative coefficient, so it cannot in general be performed by randomizing permitted tests.

Proposition 3.1 (Same algebra, different sharp permissions)**.** *Let P_0, P_1 be the pointer projections of a qubit. A fresh memory $\eta_\alpha = (\mathbf{1} + \alpha Z)/2$, $0 \le \alpha \le 1$, interacts through*

$$V = P_0 \otimes \mathbf{1} + P_1 \otimes X.$$

A Z readout of the memory implements

$$F_\alpha = \tfrac{1+\alpha}{2} P_0 + \tfrac{1-\alpha}{2} P_1.$$

For $\alpha > 0$, $W^*(F_\alpha) = W^*(P_0)$, and every hull in Definition 2.2 gives the same result for these two seeds. With arbitrary memory effects but only this single collision, the closest implementable effect to P_0 has operator-norm error $(1-\alpha)/2$. Thus equality of the ideal hulls does not give equality of their sharp-measurement permissions.

Proof. For $\alpha > 0$, $P_0 = [F_\alpha - (1-\alpha)\mathbf{1}/2]/\alpha$; each seed belongs to the algebra of the other. Paper E computes the exact single-port permission set as the parallelogram with vertices $(0,0), (1,1), ((1+\alpha)/2, (1-\alpha)/2)$ and its transpose, where (a,b) denotes $aP_0 + bP_1$ [6]. In that set $a - b \le \alpha$, so $\max(1-a, b) \ge (1-\alpha)/2$; F_α attains the bound. At $\alpha = 0$ all effects are scalar and the seed algebra is trivial. □

All nonzero gains therefore have the same ideal structure, while their finite-read abilities approach the uninformative probe continuously as $\alpha \downarrow 0$. The spectral inverse is an algebraic identity. It neither sharpens a one-shot probe nor specifies its back-action.

There is an abstract bounded coupling for an unsharp version of any self-adjoint contraction O. Put

$$\vartheta(O) = \tfrac{1}{2} \arccos(\alpha O), \qquad V_O = \exp[-i\vartheta(O) \otimes Y].$$

A pure qubit port prepared in $|0\rangle$ and read in Z yields $(\mathbf{1} + \alpha O)/2$. For a chosen gate duration t_g, $H_O = (\hbar/t_g)\vartheta(O) \otimes Y$ is bounded, and adding $\pi\hbar\mathbf{1}/(2t_g)$ makes it positive without changing the instrument. For $\alpha > 0$, this effect generates $W^*(O)$; if O is fixed by stationary noise, its spectral functions are fixed by Theorem 4.1(a). This construction specifies the required interaction, preparation and readout. It does not derive a local field Hamiltonian implementing $\vartheta(O)$, or grant that coupling to a reader whose original menu lacks it. In particular, the current and stress seeds below require a physical bounded-function readout or another justified representation of that readout.

Proposition 3.2 (Finite sharp-instrument approximation)**.** *Fix $0 < \alpha \le 1$. Let $m \ge 1$ fresh memories of polarization α undergo the preceding controlled-X collision with the* same *source qubit. Read each memory in Z, retain the majority label, and use a fair independent tie breaker if m is even. No rotation of the source occurs between collisions. Let e_m be the majority error of m independent binary tests of error $(1-\alpha)/2$. After the fine records are discarded, the coarse instrument, including its output source, is*

$$\mathcal{I}_m(\rho) = \sum_{y=0,1} |y\rangle\langle y| \otimes \sum_{z=0,1} q_m(y|z) P_z \rho P_z, \qquad q_m(z|z) = 1 - e_m, \quad q_m(1-z|z) = e_m.$$

For the sharp Lüders instrument $\mathcal{I}_{\mathrm{sh}}(\rho) = \sum_z |z\rangle\langle z| \otimes P_z \rho P_z$,

$$\tfrac{1}{2}\|\mathcal{I}_m - \mathcal{I}_{\mathrm{sh}}\|_\diamond = e_m \le e^{-m\alpha^2/2}.$$

This bound holds on arbitrary old references of the source.

Proof. The memory state is diagonal. After one collision, projection onto either memory Z outcome removes the source off-diagonal blocks: the partial maps are $\rho \mapsto \sum_z q_1(y|z) P_z \rho P_z$. Fresh memories give the product probabilities conditional on the same unchanged pointer label, as a consequence of this joint instrument. Grouping all fine outcomes by their majority gives the displayed map. For a joint input with a reference, put $R_z = (P_z \otimes \mathbf{1})\rho(P_z \otimes \mathbf{1})$. The two output-label blocks of the difference are $e_m(R_1 - R_0)$ and its negative. Their orthogonal source supports give trace norm $2e_m \operatorname{Tr} \rho$. Equivalently the map is the sharp instrument followed by a binary label flip, whose diamond distance from the identity is $2e_m$; a pointer input attains it. Hoeffding's inequality applied to the conditional $\{\pm 1\}$ samples of mean α gives the final bound. □

This is a physical finite route from an ideal pointer distinction to a task permission. To reach error $\epsilon > 0$, it suffices to use $m \geq 2\alpha^{-2} \log(1/\epsilon)$. The allowance must contain the controlled-X coupling, fresh independently prepared memories, m readouts, classical majority processing and storage; if orientation or gain is unknown, Paper E's paid references or another calibration are also needed. Discarding the fine records means restricting the task output channel; it does not physically reset their memories or erase externally retained copies. The first read destroys pointer coherence exactly as does the comparison sharp instrument. The claim concerns that instrument, not undisturbed reading of an arbitrary unknown quantum state.

Corollary 3.3 (Finite task transfer)**.** *Consider a finite adaptive circuit with arbitrary retained systems and references. Suppose its q sharp pointer-instrument slots can each be replaced by Proposition 3.2, with half-diamond errors at most e_i, uniformly over earlier records. Keep all other allowed operations the same in both circuits. Their output states have trace distance at most* $\min(1, \sum_i e_i)$*; a loss in $[0, L]$ changes in expectation by at most $L \min(1, \sum_i e_i)$.*

Proof. Include the classical history in the retained state. Replacing one slot at a time and using contractivity of all intervening physical channels bounds the full diamond distance by $2 \sum_i e_i$. Trace distance bounds the expectation of a loss in the stated interval. □

The same argument incorporates time-window or gate-error certificates at their corresponding circuit slots; Paper B gives such finite-time risk bounds for its channel class [4]. Each slot requires an instrument-level bound, not just nearby effects. A Γ-hull supplies candidate ideal observables for a task. A physical compiler supplies their actual channels and costs. The corollary connects these layers for a finite protocol meeting its conditions; it does not make all modular times, products, translations or von Neumann limits available at a fixed seed budget.

For example, applying X to the output source after a sharp pointer instrument leaves its effects unchanged but sends a P_0 input to P_1. Its output is at trace distance one from the unmodified instrument on that input. Equal probabilities, equal seed algebras and even equal sharp effects do not certify the physical task after reading.

4 Stationary sources

Three structural conditions on the fixed set are usually imposed separately: that $\mathrm{Fix}(\mathcal{E})$ be closed under products, that it be invariant under the modular group, and that it carry an ω-preserving conditional expectation. A persistent Nakajima–Zwanzig projection is a fourth. For a stationary source all of them follow.

Theorem 4.1 (Stationary source)**.** *Let the source be stationary. Then:*

(a) every $a \in \mathrm{Fix}(\mathcal{E})$ lies in the multiplicative domain of each $\mathcal{E}_t$, so $\mathcal{E}_t(axb) = a\,\mathcal{E}_t(x)\,b$ for $a, b \in \mathrm{Fix}(\mathcal{E})$, $x \in \mathcal{A}_U$, and $\mathrm{Fix}(\mathcal{E})$ is a von Neumann subalgebra;

(b) there is a unique normal conditional expectation $E_{\mathrm{fix}} : \mathcal{A}_U \to \mathrm{Fix}(\mathcal{E})$ with $\omega \circ E_{\mathrm{fix}} = \omega$; it is faithful and $E_{\mathrm{fix}} \circ \mathcal{E}_t = \mathcal{E}_t \circ E_{\mathrm{fix}} = E_{\mathrm{fix}}$;

(c) $\sigma_s(\mathrm{Fix}(\mathcal{E})) = \mathrm{Fix}(\mathcal{E})$ for all s, and the modular group of $\omega|_{\mathrm{Fix}(\mathcal{E})}$ is the restriction of σ;

(d) with $\mathcal{P} = E_{\mathrm{fix}}$ and $\mathcal{Q} = \mathrm{id} - \mathcal{P}$, $\mathcal{P}\mathcal{E}_t\mathcal{Q} = \mathcal{Q}\mathcal{E}_t\mathcal{P} = 0$ and $\mathcal{P}\mathcal{E}_t\mathcal{P} = \mathcal{P}$: the projected dynamics on the fixed algebra is the identity, and the Nakajima–Zwanzig memory kernel between the fixed algebra and its complement vanishes.

Proof. (a) For $a \in \mathrm{Fix}(\mathcal{E})$ the Kadison–Schwarz inequality gives $\mathcal{E}_t(a^*a) \geq \mathcal{E}_t(a)^*\mathcal{E}_t(a) = a^*a$, and $\omega(\mathcal{E}_t(a^*a) - a^*a) = 0$ by stationarity. Since ω is faithful, $\mathcal{E}_t(a^*a) = a^*a$; the same holds for aa^* because a^* is also fixed. By Choi's theorem [16] a belongs to the multiplicative domain of $\mathcal{E}_t$, on which $\mathcal{E}_t$ is a bimodule map. Products of fixed elements are therefore fixed, and normality of $\mathcal{E}_t$ makes the fixed set σ-weakly closed.

(b) The idea is the mean ergodic theorem of Kovács and Szűcs [24]; for semigroups of normal unital positive maps with a faithful invariant state see Watanabe [23]. By Kadison–Schwarz and stationarity, $\|\mathcal{E}_t(x)\Omega\|^2 \leq \omega(\mathcal{E}_t(x^*x)) = \|x\Omega\|^2$, so $V_t x\Omega = \mathcal{E}_t(x)\Omega$ defines a contraction semigroup on $\mathcal{H}_U$ with $V_t\Omega = \Omega$. Its Cesàro means converge strongly to the orthogonal projection P onto its fixed vectors. The means $x_T = T^{-1}\int_0^T \mathcal{E}_t(x)\,dt$ are bounded by $\|x\|$ and satisfy $x_T\Omega \to Px\Omega$; every σ-weak cluster point y has $y\Omega = Px\Omega$, and since Ω is separating the cluster point is unique, so $x_T \to y$ σ-weakly. Put $E_{\mathrm{fix}}(x) = y$. Since $\|\mathcal{E}_s(x_T) - x_T\| \leq 2s\|x\|/T$, $y \in \mathrm{Fix}(\mathcal{E})$; on $\mathrm{Fix}(\mathcal{E})$ the map is the identity, so E_{fix} is a projection of norm one onto a von Neumann algebra, hence a conditional expectation (Tomiyama). The expectation preserves the state because $\omega(E_{\mathrm{fix}}(x)) = \langle\Omega, Px\Omega\rangle = \omega(x)$, as $P\Omega = \Omega$. To prove normality, let $0 \leq x_i \uparrow x$ and $y = \sup_i E_{\mathrm{fix}}(x_i) \leq E_{\mathrm{fix}}(x)$. Normality of ω gives

$$\omega(y) = \sup_i \omega(E_{\mathrm{fix}}(x_i)) = \sup_i \omega(x_i) = \omega(x) = \omega(E_{\mathrm{fix}}(x)).$$

Faithfulness forces $y = E_{\mathrm{fix}}(x)$; thus the expectation is normal, and state preservation makes it faithful. The means of $\mathcal{E}_t(x)$ differ from those of x by at most $2t\|x\|/T$ in norm, which gives $E_{\mathrm{fix}} \circ \mathcal{E}_t = E_{\mathrm{fix}}$, and $\mathcal{E}_t \circ E_{\mathrm{fix}} = E_{\mathrm{fix}}$ because the range is fixed. A normal ω-preserving conditional expectation onto a given subalgebra is unique, since it is determined by $E(x)\Omega = e\,x\Omega$ with e the projection onto $\overline{\mathrm{Fix}(\mathcal{E})\Omega}$ [29].

(c) By Takesaki's theorem [29], a von Neumann subalgebra carries a faithful normal ω-preserving conditional expectation if and only if it is globally invariant under σ, and then σ restricts to the modular group of the restricted state.

(d) follows from (b). □

Stationarity supplies all four conditions. HAFF-E proves the fixed-algebra part [6]; the vacuum-preserving internal noise of Consciousness I [2] and the block-diagonal stationary states of HAFF-B [4] give concrete instances.

What fails without stationarity is (c). If a faithful normal state $\varphi \neq \omega$ is invariant, (a) and (b) hold with φ in place of ω and $\mathrm{Fix}(\mathcal{E})$ is invariant under the modular group of φ, which need not preserve it for ω. An entangled faithful state of $M_2 \otimes M_3$ with noise acting ergodically on the second factor is an example (check 3 of Appendix B): the fixed algebra is $M_2 \otimes \mathbf{1}$, and the modular group of the entangled state moves it.

Example 4.2 (A non-stationary qubit)**.** Let $\mathcal{A}_U = M_2$, let $\mathcal{E}_t$ multiply the off-diagonal entries in the Z-basis by $e^{-\gamma t}$, and let $\omega = \mathrm{Tr}(\rho\,\cdot)$ with ρ faithful and not diagonal. Then $\mathrm{Fix}(\mathcal{E})$ is the diagonal algebra, but $\sigma_s(P_0) = \rho^{is} P_0 \rho^{-is}$ is a rank-one projection different from $P_0 = |0\rangle\langle 0|$ for small $s \neq 0$. Two distinct nonorthogonal rank-one projections generate M_2, so the hull of the single pointer projection P_0 for any family $\Gamma \supseteq \Gamma_\sigma$ is all of M_2. For the diagonal state $\rho_\infty = \mathrm{diag}(\rho)$, which is stationary, the hull of P_0 is the two-dimensional algebra it generates.

An invariant expectation supplies a stationary partner with the same values on the persistent sector, whether or not individual trajectories relax.

Proposition 4.3 (Stationary partner)**.** *Suppose there is a faithful normal conditional expectation E_∞ of $\mathcal{A}_U$ onto* $\mathrm{Fix}(\mathcal{E})$ *with $E_\infty \circ \mathcal{E}_t = E_\infty$. Then $\omega_\infty = \omega \circ E_\infty$ is faithful, normal and stationary, agrees with ω on* $\mathrm{Fix}(\mathcal{E})$*, and Theorem 4.1 applies to $(\mathcal{A}_U, \omega_\infty, \mathcal{E})$.*

Proof. $\omega_\infty \circ \mathcal{E}_t = \omega \circ E_\infty \circ \mathcal{E}_t = \omega_\infty$; faithfulness and normality are inherited from ω and E_∞; on $\mathrm{Fix}(\mathcal{E})$, E_∞ is the identity. In standard form ω_∞ is a vector state of a cyclic and separating vector. □

For the commuting couplings of HAFF-B, E_∞ is the pinching onto the blocks; for the noises of Consciousness I it is the Haar average. The partner state defines its own modular hull. When these semigroups converge to their averages it is also the relaxed state, but Proposition 4.3 needs only the invariant expectation. The change of hull in Example 4.2 illustrates why the partner and original source must remain distinguished.

Example 4.4 (A mean without relaxation)**.** On M_2, let $\mathcal{E}_t(x) = e^{itZ} x e^{-itZ}$, with the faithful stationary source state $\tau = \mathbf{1}/2$. The fixed algebra is diagonal and its Cesàro expectation is pinching. For $\rho = (\mathbf{1} + aX)/2$, $0 < a < 1$, the Schrödinger trajectory is $e^{-itZ}\rho e^{itZ}$, whose mean tends to $\mathbf{1}/2$, but

$$\tfrac{1}{2}\|e^{-itZ}\rho e^{itZ} - \mathbf{1}/2\|_1 = a/2$$

at every time. The fixed-sector projection in Theorem 4.1 is exact; it is not a claim that the complementary dynamics decays.

Finite windows. In its commuting class, HAFF-B selects a unique largest admissible block algebra $\mathcal{B}_\pi$ when the stated window and tolerance criteria are met; smaller admissible block algebras can also exist. It proves that $\|\Phi_t^*(Y) - Y\| \leq t\,\Gamma_{\text{in}}(\pi)\|Y\|$ for $Y \in \mathcal{B}_\pi$, with Γ_{in} the block-width bound of HAFF-B [4]:

$$\Gamma_{\text{in}}(\pi) = \max_{C \in \pi} \frac{1}{2} \sum_k \gamma_k \big(\max_{x \in C} a_k(x) - \min_{x \in C} a_k(x)\big)^2.$$

The maximum is taken after summing the squared widths of each block; it is not generally the largest pairwise decoherence rate within that block. The pinching E_π onto $\mathcal{B}_\pi$ commutes with the Schur multiplier Φ_t^*, so

$$\|E_\pi \circ \Phi_t^* - E_\pi\| = \|\Phi_t^* \circ E_\pi - E_\pi\| \leq t\,\Gamma_{\text{in}}(\pi),$$

and the generator has no matrix elements between $\mathcal{B}_\pi$ and its complement. If the source state lies in $\mathcal{B}_\pi$, then $\omega \circ E_\pi = \omega$ and $\mathcal{B}_\pi$ is invariant under the modular group. On the window, the selected algebra plays the part of $\mathrm{Fix}(\mathcal{E})$ and the pinching is a Nakajima–Zwanzig projection that persists up to $t\Gamma_{\text{in}}$. In this class the memory kernel between the selected algebra and the rest vanishes exactly; a persistent record process with genuine memory needs non-commuting couplings or environments that are revisited, as in Paper F of HAFF [7].

5 Structure of the stable hull

Theorem 5.1 (The hull)**.** *Let $\Gamma \supseteq \Gamma_\sigma$ and let $S \subseteq \mathcal{A}_U$ be a seed. Write $\mathcal{H} = \mathrm{Hull}_\Gamma(S)$.*

(a) $\mathcal{H}$ is Γ-admissible and is contained in every Γ-admissible algebra that contains S. With $M_0 = W^*(S)$ and $M_{n+1} = W^*\big(M_n \cup \bigcup_{\gamma\in\Gamma} \gamma(M_n)\big)$, $\mathcal{H} = \big(\bigcup_n M_n\big)''$.

(b) There is a unique normal conditional expectation $E_\mathcal{H} : \mathcal{A}_U \to \mathcal{H}$ with $\omega \circ E_\mathcal{H} = \omega$; it is faithful. The modular group of $\omega|_\mathcal{H}$ is $\sigma|_\mathcal{H}$. The projection $e_\mathcal{H}$ onto $\overline{\mathcal{H}\Omega}$ commutes with Δ^{it} and J, and $\Delta|_{\overline{\mathcal{H}\Omega}}$, $J|_{\overline{\mathcal{H}\Omega}}$ are the modular objects of $(\mathcal{H}, \Omega)$ on that subspace.

(c) $S \subseteq S'$ implies $\mathrm{Hull}_\Gamma(S) \subseteq \mathrm{Hull}_\Gamma(S')$, and $W^*(S) \subseteq \mathcal{H}$; in particular $\mathcal{H} \neq \mathbb{C}\mathbf{1}$ as soon as S contains a nonscalar element.

Proof. (a) An intersection of admissible von Neumann algebras is admissible, and $\mathcal{A}_U$ is admissible. Every admissible algebra containing S contains each M_n by induction. Each $\gamma \in \Gamma$ is σ-weakly continuous and maps the $*$-algebra $\bigcup_n M_n$ into itself, hence maps its σ-weak closure, which is $(\bigcup_n M_n)''$, into that closure. (b) $\mathcal{H}$ is σ-invariant, so Takesaki's theorem gives $E_\mathcal{H}$ and the restriction property [29]. Since $\Delta^{it}\mathcal{H}\Omega = \sigma_t(\mathcal{H})\Omega = \mathcal{H}\Omega$, $e_\mathcal{H}$ commutes with Δ^{it}, hence with every function of Δ. For $x \in \mathcal{H}$, $Jx^*\Omega = \Delta^{1/2}x\Omega = \Delta^{1/2}e_\mathcal{H}x\Omega = e_\mathcal{H}\Delta^{1/2}x\Omega$, so J maps the dense subspace $\mathcal{H}\Omega$ of $\overline{\mathcal{H}\Omega}$ into it. The restricted operators are the modular objects of $(\mathcal{H}, \Omega)$ because the Tomita operator of $\mathcal{H}$ is the restriction of that of $\mathcal{A}_U$ to a core that is invariant under Δ^{it}. (c) is immediate from (a). □

By part (b) the hull is never an arbitrary subalgebra. It carries its own modular data, and those data are the source's. When the hull is later read for geometry, the modular flow that enters is the restriction of the flow of the source state; nothing is added.

For stable seeds of a stationary source the hull has a simple form.

Theorem 5.2 (Stable seeds)**.** *Let the source be stationary and let $S \subseteq \mathrm{Fix}(\mathcal{E})$. All statements concerning Γ_3 also assume an ambient positive translation structure and $\mathcal{E}_t \circ \mathrm{Ad}\,U(a) = \mathrm{Ad}\,U(a) \circ \mathcal{E}_t$ for $a \geq 0$.*

(a) $\mathrm{Hull}_{\Gamma_2}(S) = \mathrm{Hull}_{\Gamma_\sigma}(S) = W^*\big(\bigcup_s \sigma_s(S)\big) \subseteq \mathrm{Fix}(\mathcal{E})$. If there is an ambient Cut with the stated covariance, then $\mathrm{Hull}_{\Gamma_3}(S)$ is the smallest von Neumann algebra containing S that is invariant under σ and $\mathrm{Ad}\,U(a)$, $a \geq 0$, and it lies in $\mathrm{Fix}(\mathcal{E})$.

(b) $W^*(S) \subseteq \mathrm{Hull}(S) \subseteq \mathrm{Fix}(\mathcal{E})$ for $\mathrm{Hull} = \mathrm{Hull}_{\Gamma_2}$ or Hull_{Γ_3}. The hull is nontrivial if S contains a nonscalar element, and proper if $\mathrm{Fix}(\mathcal{E}) \neq \mathcal{A}_U$.

(c) $\mathrm{Hull}(S \cup S') = \mathrm{Hull}(S) \vee \mathrm{Hull}(S')$ for stable seeds S, S'.

(d) $E_\mathcal{H} \circ \mathcal{E}_t = \mathcal{E}_t \circ E_\mathcal{H} = E_\mathcal{H}$ for $\mathcal{H} = \mathrm{Hull}(S)$: the expectation onto the hull is a persistent projection with no back-flow.

Proof. (a) $\mathrm{Fix}(\mathcal{E})$ is a σ-invariant von Neumann algebra (Theorem 4.1) containing S, so it contains the modular hull. The noise acts as the identity on $\mathrm{Fix}(\mathcal{E})$, so the modular hull is Γ_2-admissible, and it is contained in every Γ_2-admissible algebra containing S. If $\mathcal{E}_t$ commutes with $\mathrm{Ad}\,U(a)$, then $U(a)xU(a)^*$ is fixed whenever x is, so $\mathrm{Fix}(\mathcal{E})$ is invariant under forward translations and the same argument applies. (b) follows from (a). (c) The algebra generated by two admissible subalgebras of $\mathrm{Fix}(\mathcal{E})$ is invariant under the automorphisms σ_s and the normal

endomorphisms $\mathrm{Ad}\,U(a)$, and the noise is the identity on it. (d) For $a \in \mathcal{H} \subseteq \mathrm{Fix}(\mathcal{E})$ and $x \in \mathcal{A}_U$, using bimodularity, Theorem 4.1(a) and stationarity,

$$\omega\big(a\,E_{\mathcal{H}}(\mathcal{E}_t(x))\big) = \omega(a\,\mathcal{E}_t(x)) = \omega(\mathcal{E}_t(ax)) = \omega(ax) = \omega\big(a\,E_{\mathcal{H}}(x)\big).$$

Hence $E_{\mathcal{H}}(\mathcal{E}_t(x)) - E_{\mathcal{H}}(x) \in \mathcal{H}$ is orthogonal to $\mathcal{H}\Omega$ and vanishes because Ω is separating. The second identity holds because the range of $E_{\mathcal{H}}$ is fixed. □

Part (c) answers one of the obstructions listed in HAFF-D: a completely positive map that is not multiplicative can preserve two subalgebras without preserving their join [5]. For stable seeds of a stationary source the noise is multiplicative on the relevant algebra, and the hull is a closure operator that preserves joins.

Proposition 5.3 (Records that commute with the state)**.** *Let the source be stationary and let $S \subseteq \mathrm{Fix}(\mathcal{E})$ lie in the centralizer $\mathcal{A}_U^{\omega} = \{x : \sigma_s(x) = x\ \forall s\}$. Then $\mathrm{Hull}_{\Gamma_2}(S) = W^*(S)$. In the commuting-coupling class of HAFF-B with a stationary faithful state, the block projections, which generate the record algebra, lie in the centralizer; the stable hull of any set of record projections is the abelian algebra they generate, and it contains no proper half-sided modular inclusion.*

Proof. The centralizer is a von Neumann algebra on which σ acts trivially, so $W^*(S)$ is σ-invariant; it lies in $\mathrm{Fix}(\mathcal{E})$, where the noise is trivial. In the commuting class a stationary state has vanishing entries between basis vectors with different eigenvalue tuples of the couplings [4], so it commutes with the block projections. The modular group of a faithful state on an abelian algebra is trivial, so the translation generator of a half-sided modular inclusion inside it vanishes [10] and the inclusion is an equality. □

These centralizer records do not grow under Γ_2. The thermal dephasing chain of HAFF-D at zero transverse field, where the diagonal algebra is fixed and the modular flow acts trivially on it, is an instance [5]. For a stable seed in a stationary source, Γ_2-growth requires elements moved by the modular flow. Consciousness I gives the basic example: in $M_2 \otimes M_2$ with state $\mathrm{diag}(0.8, 0.2) \otimes \frac{1}{2}\mathbf{1}$ and depolarising noise on the second factor, $X \otimes \mathbf{1}$ is fixed by the noise and its hull is $M_2 \otimes \mathbf{1}$ [2].

In finite dimension the growth is bounded by the state.

Proposition 5.4 (Finite dimension)**.** *If $\mathcal{A}_U = M_n$, $\omega = \mathrm{Tr}(\rho\,\cdot)$ is stationary and $S \subseteq \mathrm{Fix}(\mathcal{E})$, then $W^*(S) \subseteq \mathrm{Hull}_{\Gamma_2}(S) \subseteq W^*(S \cup \{\rho\}) \cap \mathrm{Fix}(\mathcal{E})$.*

Proof. $\sigma_s(x) = \rho^{is} x \rho^{-is} \in W^*(S \cup \{\rho\})$; combine with Theorem 5.2. □

For three qubits under the collective dephasing $X_1 + X_2 + X_3$ of HAFF-B, with a stationary Gibbs-type state in the block algebra and the in-block seed $X_1 - X_2$, the seed generates a 3-dimensional algebra, its hull has dimension 10, equal to that of $W^*(S \cup \{\rho\})$, and the fixed algebra has dimension 20 (check 4 of Appendix B). In type III the picture changes.

Proposition 5.5 (Chiral vacuum)**.** *Let $\mathcal{A}_U = \mathcal{A}(0, \infty)$ be the half-line algebra of a Möbius covariant net with unique vacuum Ω. Then the centralizer of ω is $\mathbb{C}\mathbf{1}$, and every σ-invariant von Neumann subalgebra other than $\mathbb{C}\mathbf{1}$ is infinite-dimensional.*

Proof. By the Bisognano–Wichmann property the modular group is the dilation group $D(t)$ [13], and $D(t)U(a)D(t)^* = U(e^{-2\pi t}a)$ for the translations. If $D(t)\xi = \xi$ for all t, then $\langle \xi, U(a)\xi\rangle = \langle \xi, U(e^{-2\pi t}a)\xi\rangle \to \|\xi\|^2$ as $t \to \infty$, so $U(a)\xi = \xi$ and ξ is a multiple of

Ω. For x in the centralizer, $x\Omega$ is dilation invariant, hence $x\Omega = c\Omega$ and $x = c\mathbf{1}$. On the orthogonal complement of Ω the representation of the translation–dilation group has positive translation generator and no invariant vectors, so it is a multiple of the unique irreducible representation of this kind, in which the dilation generator has absolutely continuous spectrum. A finite-dimensional σ-invariant subalgebra $\mathcal{N}$ would make $\mathcal{N}\Omega$ a finite-dimensional subspace invariant under Δ^{it}, hence spanned by eigenvectors of $\log\Delta$; these are multiples of Ω, so $\mathcal{N} = \mathbb{C}\mathbf{1}$. □

In a chiral vacuum every nonscalar seed is moved by the modular flow and generates an infinite-dimensional hull. Thus in this chiral vacuum the only centralizer seeds of Proposition 5.3 are scalar. Other type III source states can have nontrivial centralizers.

6 Cuts in the hull

Consciousness I needs a Cut inside the accessible algebra. A hull closed under forward translations inherits a half-sided modular inclusion from the ambient positive translation structure; it is proper when the hull is nonscalar and the invariant vector is unique.

Theorem 6.1 (Cut inheritance)**.** *Let the source carry an ambient positive translation structure U, and let $\mathcal{H} \subseteq \mathcal{A}_U$ be a von Neumann subalgebra with $\sigma_s(\mathcal{H}) = \mathcal{H}$ for all s and $U(a)\mathcal{H}U(a)^* \subseteq \mathcal{H}$ for $a \geq 0$; every Γ_3-hull has these properties. Put $\mathcal{K} = \overline{\mathcal{H}\Omega}$ and $\mathcal{N}_\mathcal{H} = U(1)\mathcal{H}U(1)^*$.*

(a) $\mathcal{K}$ reduces $U(a)$ for every $a \in \mathbb{R}$, as well as Δ^{it} and J.

(b) On $\mathcal{K}$, $(\mathcal{N}_\mathcal{H} \subset \mathcal{H}, \Omega)$ is a half-sided modular inclusion, and its translation group is $U|_\mathcal{K}$.

(c) $\mathcal{N}_\mathcal{H} \neq \mathcal{H}$ if and only if $U|_\mathcal{K} \neq \mathbf{1}$. If Ω is, up to scalars, the only translation-invariant vector, then $\mathcal{N}_\mathcal{H} \neq \mathcal{H}$ whenever $\mathcal{H} \neq \mathbb{C}\mathbf{1}$.

In particular, if Ω is, up to scalars, the only U-invariant vector, as in the vacuum representation of a chiral net, then for every seed containing a nonscalar element the Γ_3-hull contains a Cut in its vacuum-cyclic representation, and its translations are the restricted ambient translations.

Proof. (a) For $a \geq 0$ and $x \in \mathcal{H}$, $U(a)x\Omega = U(a)xU(a)^*\Omega \in \mathcal{H}\Omega$. By Theorem 5.1(b), $J\mathcal{K} = \mathcal{K}$ and $\Delta^{it}\mathcal{K} = \mathcal{K}$. Borchers' relation $U(-a) = JU(a)J$ then gives invariance under negative translations.

(b) $\mathcal{N}_\mathcal{H}\Omega = U(1)\mathcal{H}\Omega$ is dense in $U(1)\mathcal{K} = \mathcal{K}$, and Ω is separating for every subalgebra of $\mathcal{A}_U$. The modular group of $(\mathcal{H}, \Omega)$ on $\mathcal{K}$ is $\Delta^{it}|_\mathcal{K}$ by Theorem 5.1(b). For $t \geq 0$, Borchers' relation $\Delta^{-it}U(1)\Delta^{it} = U(e^{2\pi t})$ gives

$$\Delta^{-it}\mathcal{N}_\mathcal{H}\Delta^{it} = U(e^{2\pi t})\mathcal{H}U(e^{2\pi t})^* = U(1)\big[U(e^{2\pi t}-1)\mathcal{H}U(e^{2\pi t}-1)^*\big]U(1)^* \subseteq \mathcal{N}_\mathcal{H},$$

which is the half-sided condition. The modular operator of $\mathcal{N}_\mathcal{H}$ is $U(1)\Delta U(1)^*$ on $\mathcal{K}$, so $\Delta^{it}_{\mathcal{N}_\mathcal{H}}\Delta^{-it}_\mathcal{H} = U(1-e^{-2\pi t})|_\mathcal{K}$. This is the relation that defines the translation group of the inclusion in the Wiesbrock–Araki–Zsidó theorem [31, 10], which is therefore $U|_\mathcal{K}$.

(c) If $\mathcal{N}_\mathcal{H} = \mathcal{H}$, applying σ_t gives $U(b)\mathcal{H}U(b)^* = \mathcal{H}$ for all $b > 0$, hence for all b. On $\mathcal{K}$ the group U then implements automorphisms of $\mathcal{H}$ and of its commutant $\mathcal{H}' \cap \mathcal{B}(\mathcal{K})$, with positive generator and fixed vector Ω. Borchers' theorem applied to $\mathcal{H}$ and to $\mathcal{H}'$, whose modular operator is $\Delta_\mathcal{H}^{-1}$, gives both $\Delta^{it}_\mathcal{H}U(a)\Delta^{-it}_\mathcal{H} = U(e^{-2\pi t}a)$ and $\Delta^{it}_\mathcal{H}U(a)\Delta^{-it}_\mathcal{H} = U(e^{2\pi t}a)$ on $\mathcal{K}$. Hence

$U(a) = U(e^{4\pi t}a)$ for all t, and letting $t \to -\infty$ gives $U|_{\mathcal{K}} = \mathbf{1}$. Conversely, if $U|_{\mathcal{K}} = \mathbf{1}$ then $\mathcal{N}_{\mathcal{H}}$ and $\mathcal{H}$ act identically on $\mathcal{K}$. If Ω is the only invariant vector, $U|_{\mathcal{K}} = \mathbf{1}$ forces $\mathcal{K} = \mathbb{C}\Omega$ and $\mathcal{H} = \mathbb{C}\mathbf{1}$. □

In Consciousness I the Virasoro algebra B of the half-line is fixed by the noise, Ω is not cyclic for B on the ambient space, and the Cuts used for the geometry are the subalgebras $\mathrm{Vir}_1(a, \infty)$ of B on its own vacuum space [2]. Under the unique-invariant-vector hypothesis of Theorem 6.1, the Γ_3-hull of any nonscalar seed carries a Cut on its vacuum-cyclic subspace, and the modular orbit of that Cut generates the hull by the modular-orbit generation lemma of Consciousness I [2]. If the inclusion is standard, the Guido–Longo–Wiesbrock correspondence turns it into a strongly additive conformal net with translations $U|_{\mathcal{K}}$ and dilations $\Delta^{it}|_{\mathcal{K}}$ [19].

Two points limit what this achieves. The translations are the ambient ones; the hull selects a subspace and an algebra on which they act, and does not produce them. Standardness is not automatic; it holds in the symmetric examples of the next section. Since these translation-stable hulls carry the same translation and dilation groups, restricted to different subspaces, the translation–dilation group alone cannot distinguish the geometries of different seeds. An invariant of the whole triple is needed, as Consciousness I already notes [2].

7 Seeds and symmetry

This section answers three questions about the hull: whether it is unique relative to the seed, how it changes with the seed, and whether different seeds give different geometries. The answer is governed by symmetry.

7.1 A Galois law for hulls

Lemma 7.1 (Vacuum-space criterion)**.** *Let $\mathcal{B} \subseteq \mathcal{C} \subseteq \mathcal{A}_U$ be σ-invariant von Neumann algebras. Then $\mathcal{B} = \mathcal{C}$ if and only if $\overline{\mathcal{B}\Omega} = \overline{\mathcal{C}\Omega}$.*

Proof. Let $E_{\mathcal{B}}$ be the ω-preserving expectation onto $\mathcal{B}$. For $x \in \mathcal{A}_U$ and $y \in \mathcal{B}$, $\langle y\Omega, (x - E_{\mathcal{B}}(x))\Omega\rangle = \omega(y^*x) - \omega(E_{\mathcal{B}}(y^*x)) = 0$, so $e_{\mathcal{B}}x\Omega = E_{\mathcal{B}}(x)\Omega$. If the two subspaces agree and $x \in \mathcal{C}$, then $x\Omega = E_{\mathcal{B}}(x)\Omega$ and $x = E_{\mathcal{B}}(x) \in \mathcal{B}$. □

Symmetric sources. Let $\mathcal{A}_U$ be a factor with separable predual and G a compact group with a strongly continuous unitary representation u on $\mathcal{H}_U$ such that $u(g)\Omega = \Omega$ and $\alpha_g = \mathrm{Ad}\, u(g)$ maps $\mathcal{A}_U$ onto itself; if an ambient positive translation structure is present, $u(g)$ commutes with $U(a)$. Then $u(g)$ commutes with the Tomita operator, hence with Δ, and $\alpha_g \circ \sigma_s = \sigma_s \circ \alpha_g$. The action is *minimal* if $g \mapsto \alpha_g$ is injective and $(\mathcal{A}_U^G)' \cap \mathcal{A}_U = \mathbb{C}\mathbf{1}$. The *stabilizer* of a seed is the closed subgroup $\mathrm{Stab}(S) = \{g \in G : \alpha_g(x) = x \text{ for all } x \in S\}$. Izumi, Longo and Popa proved that for a minimal action of a compact group on a factor with separable predual, $K \mapsto \mathcal{A}_U^K$ is a bijection from the closed subgroups of G onto the von Neumann algebras between $\mathcal{A}_U^G$ and $\mathcal{A}_U$ [21].

Theorem 7.2 (Symmetry fixes the hull)**.** *Let G act minimally as above, let $\Gamma \supseteq \Gamma_\sigma$, and let S be a seed such that*

(i) every $\gamma \in \Gamma$ maps $\mathcal{A}_U^{\mathrm{Stab}(S)}$ into itself, and

(ii) $\mathrm{Hull}_\Gamma(S) \supseteq \mathcal{A}_U^G$.

Then $\mathrm{Hull}_\Gamma(S) = \mathcal{A}_U^{\mathrm{Stab}(S)}$. *Two seeds satisfying (i) and (ii) have the same hull if and only if they have the same stabilizer. Condition (i) holds if every* $\gamma \in \Gamma$ *commutes with every* α_g*; for a stable seed of a stationary source it also holds when* Γ *consists of* σ*, forward translations and a noise with* $\mathrm{Fix}(\mathcal{E}) = \mathcal{A}_U^{H_0}$ *for a closed subgroup* H_0.

Proof. By (i), $\mathcal{A}_U^{\mathrm{Stab}(S)}$ is Γ-admissible, and it contains S; hence $\mathrm{Hull}_\Gamma(S) \subseteq \mathcal{A}_U^{\mathrm{Stab}(S)}$. By (ii) and the Izumi–Longo–Popa theorem, $\mathrm{Hull}_\Gamma(S) = \mathcal{A}_U^{K'}$ for a closed subgroup K'. Since $S \subseteq \mathcal{A}_U^{K'}$, $K' \subseteq \mathrm{Stab}(S)$, so $\mathcal{A}_U^{\mathrm{Stab}(S)} \subseteq \mathcal{A}_U^{K'} = \mathrm{Hull}_\Gamma(S)$. The second statement is the injectivity of $K \mapsto \mathcal{A}_U^K$. For the last statement, if $S \subseteq \mathcal{A}_U^{H_0}$ then $\mathrm{Stab}(S) \supseteq H_0$, so $\mathcal{A}_U^{\mathrm{Stab}(S)} \subseteq \mathrm{Fix}(\mathcal{E})$, where the noise is the identity; σ and the translations commute with α_g. □

The hull is therefore unique relative to the seed in a strong sense: it depends on the seed only through the symmetry the seed leaves unbroken. Adding observables to a seed can only shrink the stabilizer and enlarge the hull, and observables that do not change the stabilizer add nothing.

The law has no counterpart in type I. Every automorphism of a type I factor is inner, so each α_g is implemented by a unitary $v(g) \in \mathcal{A}_U$; these unitaries commute with $\mathcal{A}_U^G$, so a nontrivial action has $(\mathcal{A}_U^G)' \cap \mathcal{A}_U \supseteq \{v(g)\}'' \neq \mathbb{C}\mathbf{1}$, and no nontrivial action is minimal. Distinct stabilizers can then give the same algebra: in M_2 with the adjoint action of $SO(3)$, both $O(2)$ and $SO(3)$ fix only the scalars (check B.1.6).

7.2 The SU(2) level-one current net

Let $\mathcal{A}$ be the $\mathrm{SU}(2)_1$ current net, $\mathcal{A}_U = M = \mathcal{A}(0,\infty)$ with the vacuum, the translations U as ambient positive translation structure and the dilations as modular group [13]. The group $SO(3)$ acts by the adjoint action, as in Assumption 4.1 of Consciousness I [2]. Write $J_n(f) = \sum_a n_a J^a(f)$ for a unit vector n and a real test function f, so that $\alpha_g(J_n(f)) = J_{R(g)n}(f)$, and $\mathcal{A}_{J_n}(I) = \{e^{iJ_n(h)} : \mathrm{supp}\, h \subset I\}''$ for the U(1) current subnet of J_n. We use four known facts.

(F1) $M^{SO(3)} = \mathrm{Vir}_1(0,\infty)$, the half-line algebra of the Virasoro net of the Sugawara stress tensor (Rehren [27]).

(F2) The Sugawara stress tensor equals the stress tensor of each current J_n: their difference commutes with J_n and is a positive-energy Virasoro field of central charge $1 - 1 = 0$, hence zero [18]. Thus $\mathrm{Vir}_1(I) \subseteq \mathcal{A}_{J_n}(I)$ [14].

(F3) On $\overline{\mathcal{A}_{J_n}(0,\infty)\Omega}$ the field J_n is a free current: the vacuum is quasi-free on $\mathcal{A}_{J_n}$, with one-particle space the completion of the real test functions in the norm $\|h\|^2 = c\int_0^\infty k\,|\hat{h}(k)|^2\,dk$, and Weyl relations $e^{iJ_n(h_1)}e^{iJ_n(h_2)} = e^{i\theta(h_1,h_2)}e^{iJ_n(h_1+h_2)}$ with real θ hold on the whole space.

(F4) Vir_1 and $\mathcal{A}$ are strongly additive [15, 32], and the fixed-point subnet of a strongly additive net under a compact group is strongly additive [32].

Proposition 7.3 (Minimality)**.** $SO(3)$ *acts minimally on* M*: the action is faithful and* $\mathrm{Vir}_1(0,\infty)' \cap M = \mathbb{C}\mathbf{1}$. *This is known [14, 32]; we include a proof.*

Proof. Let $x \in M$ commute with $\mathrm{Vir}_1(0,\infty)$. For an interval I inside the half-line, $\mathrm{Vir}_1(I) \subseteq \mathrm{Vir}_1(0,\infty)$ commutes with x; for I in the complement, $\mathrm{Vir}_1(I) \subseteq \mathcal{A}(I) \subseteq M'$ by locality. An interval that contains 0 or ∞ splits, after removing these points, into intervals of the first two kinds, and by strong additivity of Vir_1 its algebra is generated by theirs. So x commutes with

$\mathrm{Vir}_1(I)$ for every interval, hence with $V(\varphi)$ for every diffeomorphism supported in an interval, hence, since such diffeomorphisms generate $\mathrm{Diff}_+(S^1)$, with the Möbius group. Then $x\Omega$ is Möbius invariant, $x\Omega = c\Omega$ and $x = c\mathbf{1}$. Faithfulness: $\alpha_g(J_n(f)) = J_{R(g)n}(f)$ and the vectors $J^a(f)\Omega$, $a = 1, 2, 3$, are orthogonal and nonzero, so $\alpha_g = \mathrm{id}$ forces $R(g) = 1$. □

Lemma 7.4 (A current seed)**.** *For every unit vector n and every nonzero real $g \in C_c^\infty((0,\infty))$, the modular hull of $O_{n,g} = \frac{2}{\pi}\arctan J_n(g)$ is $\mathcal{A}_{J_n}(0,\infty)$.*

Proof. $\mathcal{A}_{J_n}(0,\infty)$ is invariant under dilations and contains $O_{n,g}$, so it contains the modular hull. Conversely, the hull contains $e^{itJ_n(g)}$ for all real t, and dilations act by $e^{iJ_n(h)} \mapsto e^{iJ_n(h_\lambda)}$, $h_\lambda(x) = h(x/\lambda)$. By the Weyl relations the hull contains $e^{iJ_n(h)}$ for every h in the real span L of the dilates of g; let $\mathcal{B}_L$ be the von Neumann algebra these generate. It is dilation invariant and lies in $\mathcal{A}_{J_n}(0,\infty)$.

$L + iL$ is dense in the one-particle space. With $k = e^u$ and $\psi(u) = e^u\hat{g}(e^u)$, the one-particle norm is the $L^2(\mathbb{R}, du)$ norm and a dilation is a translation of ψ. The Fourier transform $\tilde{\psi}(s) = \int_0^\infty k^{-is}\hat{g}(k)\,dk$ is the restriction to $\mathrm{Re}\, z = 1$ of the Mellin transform $\int_0^\infty k^{z-1}\hat{g}(k)\,dk$, which is analytic for $\mathrm{Re}\, z > 0$ because $\hat{g}$ is bounded and rapidly decreasing, and is not identically zero because $g \neq 0$. Its zeros on the line are isolated, so $\tilde{\psi} \neq 0$ almost everywhere, and by Wiener's theorem the translates of ψ span a dense subspace [30].

The vectors $e^{iJ_n(h)}\Omega$, $h \in L$, are multiples of the coherent vectors $\exp(ih)$. Differentiating $\exp(i\sum_j t_j h_j)$ in the real parameters t_j at zero produces all symmetric tensor products of elements of L, whose complex span is dense in each n-particle space because $L + iL$ is dense. Hence $\overline{\mathcal{B}_L\Omega}$ is the Fock space $\overline{\mathcal{A}_{J_n}(0,\infty)\Omega}$, and Lemma 7.1 gives $\mathcal{B}_L = \mathcal{A}_{J_n}(0,\infty)$. □

Lemma 7.5 (A stress seed under modular evolution)**.** *Let $0 < a < b$, $\alpha = \log a$, $\beta = \log b$, and $g \in C_c^\infty(\mathbb{R}, \mathbb{R})$ have support $[\alpha, \beta]$. Suppose $g > 0$ on (α, β), is strictly increasing near α, and strictly decreasing near β. For $f(x) = xg(\log x)$ and $O_f = \frac{2}{\pi}\arctan T(f)$,*

$$\mathrm{Hull}_{\Gamma_\sigma}(\{O_f\}) = \mathrm{Vir}_1(0,\infty).$$

The one-sided orbit $N_{O_f}^- = W^*(\sigma_s(O_f) : s \leq 0)$, restricted to the Virasoro vacuum space, is a proper standard Cut and satisfies $\mathrm{Vir}_1(b,\infty) \subseteq N_{O_f}^- \subseteq \mathrm{Vir}_1(a,\infty)$.

Proof. These are the single-orbit Cut theorem and full modular-hull corollary of Consciousness I [2]. The ambient modular group restricts to the Virasoro vacuum modular group by (F1). Every stress flow used in that construction belongs to the ambient algebra; its linear combinations, conjugations and fixed-compact-support smooth limits retain their strong closure there. The construction therefore puts $\mathrm{Vir}_1(b,\infty)$ in the one-sided orbit algebra without an independent forward-translation permission. Modular invariance of the full hull then puts $\mathrm{Vir}_1(\lambda b,\infty)$ in it for every $\lambda > 0$. These algebras generate $\mathrm{Vir}_1(0,\infty)$, which is also an upper bound because it contains the seed and is dilation invariant. Standardness and properness refer to the Virasoro vacuum space, not the larger $\mathrm{SU}(2)_1$ Hilbert space. □

Thus a current seed always has modular-only generation, while a stress seed has it under the stated bump conditions. For an arbitrary nonzero real stress smearing, the general stress-generation theorem of Consciousness I still uses dilations and forward translations. None of these closure results supplies a physical stress-flow gate from an unsharp effect.

Theorem 7.6 (Hulls in the $\mathrm{SU}(2)_1$ net)**.** *Let S be a seed containing at least one of the following:*

(i) a stress observable $O_f = \frac{2}{\pi}\arctan T(f)$ with arbitrary $0 \neq f \in C_c^\infty((0,\infty), \mathbb{R})$, with $\Gamma = \Gamma_3$;

(ii) a stress observable O_f satisfying Lemma 7.5, with $\Gamma \in \{\Gamma_\sigma, \Gamma_2, \Gamma_3\}$;

(iii) a current observable $O_{n,g}$, with $\Gamma \in \{\Gamma_\sigma, \Gamma_2, \Gamma_3\}$.

Let the noise be absent, the Haar noise $D^{SO(3)}$, or, for a stable seed, a Haar noise D^{H_0} of Consciousness I. Put $K = \mathrm{Stab}(S)$. Then:

(a) $\mathrm{Hull}_\Gamma(S) = M^K$*;*

(b) $\overline{M^K\Omega} = \mathcal{H}_U^K$*, the K-invariant vectors, and its character is*

$$\mathrm{Tr}_{\mathcal{H}_U^K} q^{L_0} = \frac{1}{\phi(q)} \sum_{j \geq 0} m_j(K)\big(q^{j^2} - q^{(j+1)^2}\big), \qquad m_j(K) = \dim V_j^K,$$

where V_j is the spin-j representation of $SO(3)$ and $\phi(q) = \prod_{n\geq 1}(1-q^n)$;

(c) $U(1)M^KU(1)^* \subset M^K$ is a standard Cut on $\mathcal{H}_U^K$, and the net it reconstructs is the fixed-point subnet $I \mapsto \mathcal{A}(I)^K$ with the restricted Möbius representation.

Proof. (a) Every Γ-admissible algebra is invariant under σ; a Γ_3-admissible algebra is also stable under forward translations. In case (i), the general stress-generation theorem of Consciousness I [2] puts $\mathrm{Vir}_1(0,\infty)$ in the hull. In case (ii), Lemma 7.5 gives the same inclusion using only modular evolution. In case (iii), it follows from Lemma 7.4 and (F2). By (F1) this is $M^{SO(3)}$. The modular group and the translations commute with α_g; the Haar noise $D^{SO(3)}$ commutes with every α_g by invariance of Haar measure; a noise D^{H_0} has fixed algebra M^{H_0} [2]. Proposition 7.3 and Theorem 7.2 give the claim.

(b) Since $u(k)\Omega = \Omega$, the projection $P_K = \int_K u(k)\,dk$ onto $\mathcal{H}_U^K$ maps $x\Omega$ to $Q_K(x)\Omega \in M^K\Omega$, where Q_K is the Haar average; as $M\Omega$ is dense, $M^K\Omega$ is dense in $\mathcal{H}_U^K$. The vacuum module decomposes under $SO(3) \times \mathrm{Vir}$ as $\bigoplus_{j\geq 0} V_j \otimes L(1, j^2)$, and the character of the irreducible $c = 1$ module $L(1, j^2)$ is $q^{j^2}(1 - q^{2j+1})/\phi(q)$ [27, 28, 22]; taking K-invariants gives the formula.

(c) $\mathcal{A}^K$ is a Möbius covariant subnet, strongly additive by (F4), and Ω is cyclic on $\mathcal{H}_U^K$ for each $\mathcal{A}(I)^K$ by the Reeh–Schlieder property of the subnet in its vacuum representation. Since $M^K = \mathcal{A}^K(0,\infty)$ and $U(1)M^KU(1)^* = \mathcal{A}^K(1,\infty)$, the relative commutant contains $\mathcal{A}^K(0,1)$, so the Cut is standard, and the Guido–Longo–Wiesbrock reconstruction returns $\mathcal{A}^K$ [19]. The Möbius representation of a Möbius covariant net is fixed by the modular groups of its interval algebras [13], so it is the restriction of the ambient one. □

Theorem 7.7 (Seeds are distinguished by geometry)**.** *For two seeds S_1, S_2 as in Theorem 7.6, the Cut triples $(U(1)\mathcal{H}_iU(1)^* \subset \mathcal{H}_i, \Omega)$ on $\overline{\mathcal{H}_i\Omega}$, $\mathcal{H}_i = \mathrm{Hull}_\Gamma(S_i)$, are isomorphic if and only if* $\mathrm{Stab}(S_1)$ *and* $\mathrm{Stab}(S_2)$ *are conjugate in $SO(3)$.*

Proof. If $\mathrm{Stab}(S_2) = g\,\mathrm{Stab}(S_1)g^{-1}$, then $u(g)$ maps $M^{\mathrm{Stab}(S_1)}$ onto $M^{\mathrm{Stab}(S_2)}$, commutes with U and Δ, and fixes Ω. Conversely, an isomorphism of standard triples induces an isomorphism of the reconstructed nets intertwining their Möbius representations [19, 13], so the two characters of Theorem 7.6(b) agree. Multiplying by $\phi(q)$ gives $\sum_j (m_j - m_{j-1})q^{j^2}$, from which every $m_j(K)$ is read off. With $\chi_j(\theta) = \sum_{m=-j}^{j} e^{im\theta}$ the character of V_j at rotation angle θ, $m_j - m_{j-1} = \int(\chi_j - \chi_{j-1})\,d\mu_K = 2\int \cos(j\theta)\,d\mu_K(\theta)$ for $j \geq 1$, where μ_K is the image of Haar measure of K under the rotation angle $\theta \in [0,\pi]$. The cosine moments determine μ_K, and μ_K determines the conjugacy class of K (Lemma A.1). □

The classified triples use the ambient-translation inclusion $U(1)M^K U(1)^* \subset M^K$. The seed's one-sided orbit $N_{O_f}^- \subset \mathrm{Vir}_1(0,\infty)$ in Lemma 7.5 is a different standard inclusion. Its reconstructed translations need not equal U. Consciousness I's Conjecture T1(b) asserts that this one-sided orbit is not an ordinary translated half-line; that noncoincidence remains open. The character invariant here is proved for the ambient-translation triples.

The identification compares the entire standard Cut triples and their conformal subnets on the same ambient light ray. Their nonisomorphism does not assert that their underlying spacetime metrics differ. The whole character is needed: the multiplicities of C_n are $2\lfloor j/n \rfloor + 1$, so its hull and the U(1) hull have the same character below level n^2. A reader resolving finitely many conformal levels distinguishes only stabilizers whose characters already differ below that level.

Table 1 lists the first coefficients. Some symmetries show up late: the hulls of seeds with octahedral and icosahedral stabilizers agree with the Virasoro hull up to level 15, and the icosahedral hull differs from the Virasoro hull first at level 36, where the first invariant harmonic of degree six appears.

stabilizer K	$m_j(K)$	character coefficients, levels 0–12
$SO(3)$	δ_{j0}	1, 0, 1, 1, 2, 2, 4, 4, 7, 8, 12, 14, 21
$O(2)$	1 for even j, else 0	1, 0, 1, 1, 3, 3, 6, 7, 12, 14, 22, 27, 40
$SO(2)$	1	1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77
D_2		1, 0, 1, 1, 4, 4, 8, 10, 17, 21, 33, 42, 62
T		1, 0, 1, 1, 2, 2, 4, 4, 7, 9, 13, 16, 24
C_2	$2\lfloor j/2 \rfloor + 1$	1, 1, 2, 3, 7, 9, 15, 21, 32, 44, 64, 86, 121
$\{1\}$	$2j+1$	1, 3, 4, 7, 13, 19, 29, 43, 62, 90, 126, 174, 239

Table 1: Vacuum characters of the hulls M^K in the SU(2)$_1$ net. The rows for $SO(3)$, $SO(2)$ and $\{1\}$ are the Virasoro, U(1) current and SU(2)$_1$ vacuum characters; the $O(2)$ row counts partitions into an even number of parts. Computed in check B.2 of Appendix B.

Corollary 7.8 (Dephasing about one axis)**.** *Let the noise be D^{H_z}, with H_z the rotations about e_3. The seed and operation family in the hull statements below satisfy Theorem 7.6: arbitrary stress smearings use Γ_3; the stated log-bump stress seeds and current seeds also admit Γ_σ and Γ_2.*

(a) $\mathrm{Fix}(D^{H_z}) = M^{H_z} = \mathcal{A}_{J_{e_3}}(0,\infty)$*, the U(1) current algebra of the half-line.*

(b) A current observable $O_{n,g}$ is stable if and only if $n = \pm e_3$. Every stable seed that contains a stress or a current observable has one of exactly three hulls: $\mathcal{A}_{J_{e_3}}(0,\infty)$, its $\mathbb{Z}_2$-orbifold $M^{O(2)_z}$ under $J_{e_3} \mapsto -J_{e_3}$, or $\mathrm{Vir}_1(0,\infty)$. They are attained by $\{O_{e_3,g}\}$, by $\{O_f, x\}$ with $x \in M^{O(2)_z} \setminus \mathrm{Vir}_1(0,\infty)$, and by $\{O_f\}$, and their characters differ at levels 1 and 4.

(c) Under the Haar noise $D^{SO(3)}$ every stable seed containing a stress observable has the hull $\mathrm{Vir}_1(0,\infty)$. Without noise every conjugacy class of closed subgroups occurs as a stabilizer, and two current observables along non-parallel axes generate all of M under the modular flow alone.

Proof. (a) $\mathrm{Fix}(D^{H_z}) = M^{H_z}$ [2]. Since $J_n(g)\Omega = \sum_a n_a J^a(g)\Omega$ with orthogonal nonzero vectors $J^a(g)\Omega$, α_g fixes $O_{e_3,g}$ exactly when $R(g)e_3 = e_3$; so $\mathrm{Stab}(O_{e_3,g}) = H_z$, and Lemma 7.4 with Theorem 7.6 gives $\mathcal{A}_{J_{e_3}}(0,\infty) = M^{H_z}$. (b) The same argument shows that $O_{n,g} \in M^{H_z}$

only for $n = \pm e_3$. A stable seed has $\mathrm{Stab}(S) \supseteq H_z$. The closed subgroups of $SO(3)$ containing H_z are H_z, its normalizer $O(2)_z$ and $SO(3)$: the identity component of such a subgroup is H_z or $SO(3)$, and in the first case the subgroup normalizes H_z. The rotation by π about e_1 acts on M^{H_z} by $J_{e_3} \mapsto -J_{e_3}$, so $M^{O(2)_z}$ is the orbifold. An element $x \in M^{O(2)_z}$ outside $\mathrm{Vir}_1(0,\infty)$ exists by the Galois correspondence, and $\mathrm{Stab}(\{O_f, x\})$ contains $O(2)_z$ and is not $SO(3)$, so it equals $O(2)_z$. The characters follow from $m_j(H_z) = 1$, $m_j(O(2)_z) = 1$ for even j and 0 otherwise, and $m_j(SO(3)) = \delta_{j0}$. (c) The Haar noise fixes exactly $M^{SO(3)}$. Without noise, the self-adjoint part of M^K is a seed with stabilizer K by the Galois correspondence, and $\mathrm{Stab}(\{O_{n,g}, O_{n',g'}\})$ is trivial for non-parallel n, n'. □

Under axis dephasing, stable seeds in this stress/current-containing class therefore have exactly three ideal hulls, and they are told apart by the spectrum of the conformal Hamiltonian at the first and fourth levels. A reader whose ideal seed contains a bounded injective function of $J_{e_3}(g)$ has the whole fixed algebra as its hull; a reader that registers only quantities insensitive to the sign of J_{e_3}, together with the energy density, gets the orbifold; a reader that registers only energy gets the Virasoro algebra.

Lifetime of a hull. Real noise breaks a symmetry slightly. Let $\mathcal{E}_t = D_t^{H_z} \circ D_{st}^{SO(3)}$, with rates κ and $s\kappa$, $0 < s \ll 1$. The two semigroups commute, because $Q_{SO(3)}Q_{H_z} = Q_{H_z}Q_{SO(3)} = Q_{SO(3)}$, and the fixed algebra of the product is $\mathrm{Vir}_1(0,\infty)$ [2].

Proposition 7.9 (A plateau of length $1/s$)**.** *For every closed $K \supseteq H_z$ with $M^K \neq \mathrm{Vir}_1(0,\infty)$,*

$$\sup_{x \in M^K,\ \|x\| \le 1} \|\mathcal{E}_t(x) - x\| = c_K\big(1 - e^{-s\kappa t}\big), \qquad 1 \le c_K \le 2.$$

Thus M^K is ε-stable on $[0,T]$ in the sense of HAFF-B exactly when $c_K(1 - e^{-s\kappa T}) \le \varepsilon$, which for small ε is a window of length $\varepsilon/(c_K s\kappa)$.

Proof. On M^{H_z} the first factor is the identity and $D_{st}^{SO(3)}(x) = e^{-s\kappa t}x + (1 - e^{-s\kappa t})Q_{SO(3)}(x)$, so $\mathcal{E}_t(x) - x = -(1 - e^{-s\kappa t})(x - Q_{SO(3)}(x))$. Put $c_K = \sup\{\|x - Q_{SO(3)}(x)\| : x \in M^K, \|x\| \le 1\} \le 2$. For $y \in M^K \setminus \mathrm{Vir}_1(0,\infty)$ the element $x = (y - Q_{SO(3)}y)/\|y - Q_{SO(3)}y\|$ has $Q_{SO(3)}(x) = 0$, so $c_K \ge 1$. □

The U(1) hull of a current reader thus lives for a time inversely proportional to the strength of the symmetry-breaking noise, after which only the Virasoro algebra remains stable. This is the type III counterpart of the plateau that HAFF-B computes for the GHZ geometry [4]. A possible maintenance mechanism would correct the symmetry-breaking jumps on their characteristic timescale $(s\kappa)^{-1}$. The plateau formula establishes the passive lifetime; it supplies neither a universal correction-rate bound nor the energy cost of maintaining this type III hull. Q-RAIF B gives a field-coupled correcting actuator in a finite model [9]; extending it to the present hull is a concrete test of HAFF-D's conjecture that maintained access carries a gravitating stress-energy flux [5].

8 Ideal seeds and finite reading budgets

The symmetry classification concerns ideal operator seeds. A finite processing budget first restricts a physical reader's instrument menu; it does not assign a cost to the spectral closure of a current field. Appendix C gives a separate finite-coordinate coding model. No dictionary from its coordinate projections to the current or stress observables is supplied here.

Corollary 8.1 (A single ideal current seed)**.** *Under the hypotheses of Corollary 7.8, the ideal spectral seed* $\{O_{e_3,g}\}$*,* $g \neq 0$*, has the whole fixed algebra* M^{H_z} *as its hull. No stable seed has a larger hull. Without noise, two ideal seeds containing bounded injective functions of nonparallel current components have* M *as their hull.*

Proof. Lemma 7.4 and Theorem 7.6 identify the one-axis hull. Theorem 5.2 bounds every stable hull by the fixed algebra. Two nonparallel current components have trivial common stabilizer, so Theorem 7.6(a) gives M. □

This corollary uses ideal bounded functional calculus, all modular times and von Neumann closure. Proposition 3.2 instead prices an actual finite pointer instrument. Neither statement converts a one-coordinate processing allowance into permission to implement the whole current algebra. When a physical menu implements a specified spectral seed, its ideal hull depends on the stabilizer of that seed; the menu and its finite cost remain separate inputs.

9 How a hull responds to excitations

HAFF-D connects a selected algebra to gravity through the response of its modular data to excitations of the state [5], and Consciousness I computes the relative entropy lost when a Cut is displaced [2]. Whether these responses depend on the seed is decided by the following facts. For a normal state ψ and a von Neumann subalgebra $\mathcal{N}$, write $S_{\mathcal{N}}(\psi\|\omega)$ for Araki's relative entropy of the restrictions, and $\omega_W = \omega(W^* \cdot W)$ for a unitary W.

Proposition 9.1 (Response and the hull)**.** *Let* $\mathcal{H} \subseteq \mathcal{H}'$ *be* σ*-invariant von Neumann subalgebras.*

(a) For a unitary $W \in \mathcal{H}$*,* $S_{\mathcal{H}}(\omega_W\|\omega) = S_{\mathcal{A}_U}(\omega_W\|\omega)$*, and the same holds with the arguments exchanged.*

(b) For every normal state ψ*,* $S_{\mathcal{H}}(\psi\|\omega) \leq S_{\mathcal{H}'}(\psi\|\omega)$*. If the right side is finite, equality holds if and only if* $\psi|_{\mathcal{H}'} = \psi|_{\mathcal{H}} \circ E_{\mathcal{H}}|_{\mathcal{H}'}$*.*

Proof. (a) Since $W \in \mathcal{H}$ and $E_{\mathcal{H}}$ is $\mathcal{H}$-bimodular and ω-preserving, $\omega_W \circ E_{\mathcal{H}} = \omega_W$ and $\omega \circ E_{\mathcal{H}} = \omega$. Relative entropy is monotone under restriction to $\mathcal{H}$ and under the channel $\varphi \mapsto \varphi \circ E_{\mathcal{H}}$, which maps the two restrictions back to the two states, so the two values coincide. (b) The inequality is monotonicity under restriction. The equality condition is Petz's characterization of sufficient subalgebras [26, 25]: with $\omega \circ E_{\mathcal{H}} = \omega$, the recovery map for ω is $E_{\mathcal{H}}$ itself. □

Corollary 9.2 (Stress excitations do not see the seed; current excitations do)**.** *In the SU(2)*$_1$ *net:*

(a) for $W = e^{iT(f)}$ *with* $f \in C_c^\infty((0,\infty),\mathbb{R})$*, every hull* M^K *gives the same relative entropy, so the coherent-state formula of Hollands and the Cut-displacement theorem of Consciousness I [20, 2] hold verbatim in every hull of Theorem 7.6;*

(b) for $W = e^{iJ_{e_3}(h)}$ *with* $0 \neq h \in C_c^\infty((0,\infty),\mathbb{R})$*,*

$$S_{\mathrm{Vir}_1(0,\infty)}(\omega_W\|\omega) < S_{\mathcal{A}_{J_{e_3}}(0,\infty)}(\omega_W\|\omega) = 2\pi\langle W\Omega, KW\Omega\rangle,$$

where $K = -\frac{1}{2\pi}\log\Delta$ *is the dilation generator.*

Proof. (a) $e^{iT(f)} \in \mathrm{Vir}_1(0,\infty) \subseteq M^K$; apply Proposition 9.1(a). (b) $W \in \mathcal{A}_{J_{e_3}}(0,\infty)$, so the middle term equals the ambient value, which for a unitary of M is $-\langle W\Omega, \log \Delta\, W\Omega\rangle$; it is finite because the coherent state has finite dilation energy. On $\mathcal{H}' = M^{H_z}$ the expectation onto $\mathrm{Vir}_1(0,\infty)$ is the Haar average $Q = Q_{SO(3)}$. For a real $k \in C_c^\infty((0,\infty))$ and the odd function arctan, the rotation by π about e_1 maps $\arctan J_{e_3}(k)$ to its negative and leaves Q unchanged, so $Q(\arctan J_{e_3}(k)) = 0$. In the state ω_W, $J_{e_3}(k)$ has the vacuum law, a centred Gaussian, shifted by a real number μ proportional to $\int hk'\,dx$, because the commutator of the current is a multiple of the identity; $\mu \neq 0$ for $k = h'$. Since arctan is odd and strictly increasing, $\omega_W(\arctan J_{e_3}(k)) \neq 0 = \omega_W(Q(\arctan J_{e_3}(k)))$. By Proposition 9.1(b) the inequality is strict. □

For a stress excitation lying in every hull, its relative entropy and modular-energy response agree across the hulls, by Proposition 9.1(a). The modular operators are restrictions of the ambient operator to different vacuum-cyclic spaces. The equality applies to excitations lying in every hull; arbitrary states can have different relative entropies on those spaces. Relative entropy can differ: the hull of an energy reader underestimates the information in a current excitation that the hull of a current reader sees in full. Whatever role a selected algebra plays in a gravitational response, its dependence on the seed enters through excitations outside the Virasoro algebra; stress excitations alone cannot detect which seed was selected. In higher dimensions Consciousness II shows that selected algebras with the required stress and boundary readings, on a conformal net in Minkowski space and conditional on its supplied MC_η ball-consistency law and source regularity, give the same matter-sourced linearized Einstein tensor when they use one source and normalization; their relative entropies can differ [1]. That paper also constructs an exact neutral scalar preparation inside a proper stable hull: a seed containing the stress readings and one invariant quadratic observable contains a local unitary that produces nonzero Fourier weight. With permission to implement this pulse, the observer can test the intrinsic response without a charged-field phase reference. This construction is in the higher-dimensional matter theory; the chiral Cut classification here does not supply its spacetime or its modular-consistency law.

10 What is still supplied, and three conjectures

The construction derives the fixed algebra, its expectation and its modular invariance from stationarity; the hull and its expectation from a seed; a proper Cut in every nonscalar translation-stable hull from an ambient positive translation structure with a unique invariant vector; and, under the hypotheses of Theorem 7.2, the hull from the seed's stabilizer. The following remain inputs.

(1) The ambient algebra and its state. The Cut-inheritance theorem additionally supplies positive translations; the log-bump construction instead uses a one-sided modular orbit inside a given chiral net. Neither derives that ambient net. Two nested hulls invariant under the same ambient modular group have a two-sided invariant inclusion, not a proper Cut of the kind used here. A one-sided construction is needed within the selected algebra.

(2) The coupling to the environment. HAFF-B derives the selected algebra from the couplings, and Consciousness I from the noise group; neither derives the couplings.

(3) Stationarity, or a stationary partner supplied by an invariant expectation (Proposition 4.3). Relaxation requires an additional dynamical statement (Example 4.4).

(4) The task that fixes the foreground. T-DOME II gives the optimal reading directions for a given task; the task is supplied.

(5) A minimal compact symmetry for the classification. In the $\mathrm{SU}(2)_1$ net it is verified; elsewhere the hull is still determined by Theorems 5.1 and 5.2, but its identification with a fixed-point algebra is open.

Conjecture 10.1 (Quantum symmetry fixes the hull)**.** *Let $\mathcal{A}$ be a completely rational conformal net and $\mathcal{B} \subset \mathcal{A}$ a Möbius covariant subnet of finite index with $\mathcal{B}(I)' \cap \mathcal{A}(I) = \mathbb{C}\mathbf{1}$. For every seed whose Γ_3-hull contains $\mathcal{B}(0,\infty)$, the hull is the half-line algebra of the intermediate net fixed by the generalized stabilizer of the seed in the quantum symmetry of $\mathcal{B} \subset \mathcal{A}$, and its vacuum character is the corresponding partial sum of branching functions.*

The mechanism is Theorem 7.2 with the compact group replaced by the symmetry that a finite-index inclusion carries. Support comes from the compact case proved here, from the extension of the Izumi–Longo–Popa correspondence to compact Kac algebras [21], and from the correspondence between intermediate extensions and intermediate Q-systems [11]. The strongest objection is that a Γ_3-hull is invariant under dilations and forward translations but is not known in general to be the half-line algebra of a local subnet; in the $\mathrm{SU}(2)_1$ case this follows from the Galois law, not from the hull construction. Without the trivial relative commutant the conjecture fails, since intermediate algebras then need not be fixed by any symmetry.

Conjecture 10.2 (Window selection in type III)**.** *For the noise $D_t^{H_z} \circ D_{st}^{SO(3)}$ and seeds containing a stress observable, a window with $s\kappa T \ll \varepsilon \ll \kappa T$ selects M^{H_z} as the unique largest stable hull in this class: it is ε-stable on $[0,T]$, and no hull M^K with $K \not\supseteq H_z$ is. As the window grows past $\varepsilon/(s\kappa)$ the selected hull drops to $\mathrm{Vir}_1(0,\infty)$, and the step is visible in the vacuum character.*

This is HAFF-B's window theorem [4] in type III. Proposition 7.9 proves the stability half. For $K \subsetneq H_z$, an element $x \in M^K$ with $Q_{H_z}(x) = 0$ and $\|x\| = 1$ has $\|\mathcal{E}_t(x) - x\| = 1 - e^{-(1+s)\kappa t}$, which proves the instability half for those K. The objection is that for other K the supremum of the instability over the unit ball of M^K has no evident lower bound independent of K; localized charged unitaries are the natural candidates for proving one.

Conjecture 10.3 (Every hull contains the Virasoro algebra)**.** *In the $\mathrm{SU}(2)_1$ net every nonscalar Γ_3-hull contains $\mathrm{Vir}_1(0,\infty)$. Then Theorem 7.6 classifies all hulls of the net.*

The mechanism is Theorem 6.1: a nonscalar hull carries a Cut, and if the Cut is standard it reconstructs a conformal subnet $\mathcal{B}$. If $\mathcal{B}$ is diffeomorphism covariant with central charge one, the coset of $\mathcal{B}$ in $\mathcal{A}$ has central charge zero and is trivial, so the stress tensor of $\mathcal{B}$ is the Sugawara tensor and $\mathrm{Vir}_1 \subseteq \mathcal{B}$ [18]. A Möbius covariant subnet of central charge below one cannot occur: Carpi showed that every Möbius covariant subsystem of the $\mathrm{SU}(2)_1$ net is a fixed-point net $\mathcal{A}^H$ for a closed subgroup $H \subseteq SO(3)$, and all of these contain Vir_1 [14]. The conjecture therefore reduces to a lifting problem, which is the objection: the net reconstructed from a standard hull Cut must be shown to be a conformal subsystem of the ambient net, that is, its local algebras $\mathcal{H} \cap (U(1)\mathcal{H}U(1)^*)'$ must lie in the ambient local algebras and its Möbius representation must be the restriction of the ambient one. Standardness of the inherited Cut is also not automatic.

11 Conclusion

The physical reader and the ideal closure carry different information. A nonzero-gain probe can generate a sharp pointer algebra without being able to read that pointer sharply in one shot. In the explicit nondemolition model, repetition with fresh ports makes the ideal instrument available to any prescribed finite accuracy, with a budget depending on gain. The instrument-level estimate survives old references and finite adaptive composition. This is the task bridge; spectral closure alone cannot supply it.

Stationarity of the source makes the fixed algebra a modular-invariant expected subalgebra and supplies a persistent projection with no memory kernel, without requiring relaxation of every trajectory. The stable hull of a seed fixed pointwise by the noise carries the source's own modular data and lies between the seed and the fixed algebra. A nonscalar hull that is also closed under forward translations carries a proper Cut on its vacuum space when the ambient positive translation group leaves only the vacuum invariant. Translation covariance of the noise keeps this larger hull in the fixed algebra. Under a minimal compact symmetry, a hull containing the symmetry-fixed algebra is fixed by the stabilizer of the seed when its closure maps preserve the stabilizer-fixed algebra. Within this class the accessible algebra is unique relative to what is read and changes only when the symmetry left unbroken by the seed changes. In the $\mathrm{SU}(2)_1$ net the vacuum characters of the hulls tell the stabilizers apart; a single current reading generates the U(1) current algebra under the modular flow. For the positive log-bump stress class, modular flow alone generates the Virasoro half-line algebra, and the one-sided orbit constructs a proper standard Cut on its own vacuum space. In the stated stress/current seed class, dephasing about one axis leaves exactly three ideal hulls, and each lasts as long as the noise respects its symmetry. The response of a hull to stress excitations does not depend on the seed, while its response to a current excitation does.

The source–access–Cut chain has a derived net-theoretic middle. The retained seed fixes the ideal algebra through symmetry in the classified cases; physical protocols decide which distinctions can be used within a task budget. The translation-stable algebra inherits the source's modular data and a Cut. In the $\mathrm{SU}(2)_1$ realization that Cut is standard and reconstructs its conformal subnet on the inherited light ray. Different seeds can select nonisomorphic nets there without producing distinct spacetime metrics. The ambient net and physical reader remain inputs. The log-bump construction removes an independent forward-translation requirement from ideal stress-seed generation, but still presupposes the ambient stress tensor and modular evolution. Coupling, preparation permissions and the task remain physical inputs. Learned access and retained-history mechanisms can change these inputs during an actual run; the hull theorem then identifies the structure supported by the resulting seed. It does not by itself explain the origin of the first available coupling.

A Rotation angles and conjugacy classes

Lemma A.1. *For a closed subgroup $K \subseteq SO(3)$ let μ_K be the distribution of the rotation angle $\theta \in [0, \pi]$ under normalized Haar measure. Then μ_K determines K up to conjugacy.*

Proof. Up to conjugacy the closed subgroups are $SO(3)$, $O(2)$, $SO(2)$, the icosahedral, octahedral and tetrahedral groups I, O, T of orders $60, 24, 12$, the dihedral groups D_n of order $2n$, $n \geq 2$, and the cyclic groups C_n, $n \geq 1$. The three infinite groups have μ_K with density $(1 - \cos\theta)/\pi$, the density $1/\pi$, and half the density $1/\pi$ plus an atom $\frac{1}{2}$ at π; these correspond respectively to $SO(3)$, $SO(2)$, and $O(2)$. The finite ones are atomic. For finite K the atom at

0 is $1/|K|$, which fixes the order. Among groups of equal order, the atom at π separates them: it is $1/n$ for C_n with n even and 0 for n odd; $(n+1)/2n$ for D_n with n even and $\frac{1}{2}$ for n odd; $\frac{1}{4}$ for T, $\frac{3}{8}$ for O and $\frac{1}{4}$ for I. For order $2m$, C_{2m} has $1/2m$ and D_m at least $\frac{1}{2}$, equal only for $m = 1$, where D_1 is conjugate to C_2. For orders 12, 24 and 60 the three values are $(\frac{1}{12}, \frac{7}{12}, \frac{1}{4})$, $(\frac{1}{24}, \frac{13}{24}, \frac{3}{8})$ and $(\frac{1}{60}, \frac{31}{60}, \frac{1}{4})$ for C, D, and T, O, I respectively. □

B Numerical checks

The supplementary source map identifies the following diagnostics and their retained outputs. Finite samples test the stated identities; the infinite-dimensional conclusions use the proofs in the text.

B.0 Physical permission bridge. The controlled-X complete instrument is checked on an old Bell reference and eleven random mixed inputs at $\alpha = 0, .1, .6, 1$ and $m = 1, 2, 3, 5$, including the fair tie rule and post-read-flip counterexample. The sharp half-diamond identity is analytic (Proposition 3.2).

B.1 Finite hulls. Closure under products, adjoints, the modular generator and noise reproduces the nonstationary qubit hull M_2, its pinched-state two-dimensional hull, and the $M_2 \otimes 1$ stationary fixed sector. The three-qubit example has seed, hull and fixed dimensions $3, 10, 20$, respectively. Other checks distinguish equal type-I fixed algebras with different stabilizers, verify persistence and joins, and test Proposition C.2 on 200 task matrices.

B.2 Characters. The class data of C_n, D_n for $n \leq 40$, the three finite polyhedral groups and $SO(2), O(2), SO(3)$ give 85 distinct multiplicity sequences through $j = 60$. Their characters reproduce the table and the Virasoro, current, SU(2) level-one and even-Fock low-energy coefficients. Finite-level agreement does not replace the analytic character and rotation-angle arguments.

C A separate finite-coordinate coding model

Here V is an abstract finite-dimensional task coordinate space. The propositions do not construct a net observable from a coordinate projection and are not premises of Corollary 8.1. T-DOME II supplies the Gaussian task optimizer used in this model.

Centred and uncentred task losses. A rank optimizer for centred omission loss need not optimize an uncentred squared loss: the latter also contains $(\sum_{i \notin I} \mathbb{E}\lambda_i)^2$. For three independent channels with means and variances $(0, 1), (1, .1), (1, .1)$ and one retained channel, the uncentred minimizer keeps a channel with smaller centred weight. The coding result in Appendix C therefore retains its stated task assumptions.

Proposition C.1 (Rank under a processing budget)**.** *Let a reader hold a real coordinate space V of dimension D, let admissible encoders be orthogonal projections Q on V with processing cost $h_\mu \operatorname{rank} Q$, $h_\mu > 0$, and assume:*

(1) the full-rank encoder is infeasible, $h_\mu D > \dot{C}_{\text{budget}}$;

(2) the distortion functional attains its minimum on the feasible projections and strictly decreases whenever a feasible projection is enlarged by one coordinate;

(3) a group G acts irreducibly and orthogonally on V, the distortion is G-invariant, and $0 < \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$.

Then every minimizing encoder has rank $k^* = \lfloor \dot{\mathcal{C}}_{\text{budget}}/h_\mu \rfloor < D$, its range is a k^*-dimensional foreground, and no such projection is G-invariant, so choosing a minimizer breaks the G-symmetry.

Proof. Feasibility gives rank $Q \leq k^*$, and strict improvement excludes smaller rank. A minimizing projection that commuted with the irreducible representation would have invariant range and kernel, hence would be 0 or $\mathbf{1}$, contrary to $0 < k^* < D$. □

The rank charge, strict improvement and irreducible action are inputs of this coding model. They are not consequences of persistence, and the proposition does not say that symmetry breaking is unavoidable outside this class of encoders.

Which k^* directions are kept is answered by T-DOME II. For a budgeted reader whose task is linear in a fresh Gaussian input and whose loss is quadratic, with all rank-k linear projectors allowed, the best risk of a rank-k reading policy is attained by the top-k eigenspace of the posterior task matrix $\widehat{W}_M$ built from its memory M, and the optimal subspace is unique when $\lambda_k(\widehat{W}_M) > \lambda_{k+1}(\widehat{W}_M)$ [8]. These directions are record channels when $\widehat{W}_M$ commutes with the projections onto the record channels, and approximately so otherwise.

Proposition C.2 (Foreground and record channels)**.** *Let the reader's D channels be record channels, and write $\widehat{W}_M = W_{\text{d}} + W_{\text{o}}$ with W_{d} diagonal in the channel basis and W_{o} its off-diagonal part. Let $\Delta_k > 0$ be the gap between the k-th and $(k+1)$-st largest diagonal entries and P_{ch} the projection onto the corresponding k channels. If $W_{\text{o}} = 0$, the optimal foreground is spanned by those k channels. If $\|W_{\text{o}}\| < \Delta_k$, any rank-$k$ optimal projection P_{opt} satisfies*

$$\|P_{\text{opt}} - P_{\text{ch}}\| \leq \frac{\|W_{\text{o}}\|}{\Delta_k - \|W_{\text{o}}\|}.$$

Proof. The first statement is the capability theorem of T-DOME II for a diagonal matrix. For the second, Weyl's inequality separates the top k eigenvalues of $\widehat{W}_M$ from the remaining diagonal entries of W_{d} by at least $\Delta_k - \|W_{\text{o}}\|$, and the Davis–Kahan $\sin\Theta$ theorem [17] bounds the sine of the largest principal angle, which for projections of equal rank equals $\|P_{\text{opt}} - P_{\text{ch}}\|$, by $\|W_{\text{o}}\|$ divided by that separation. □

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